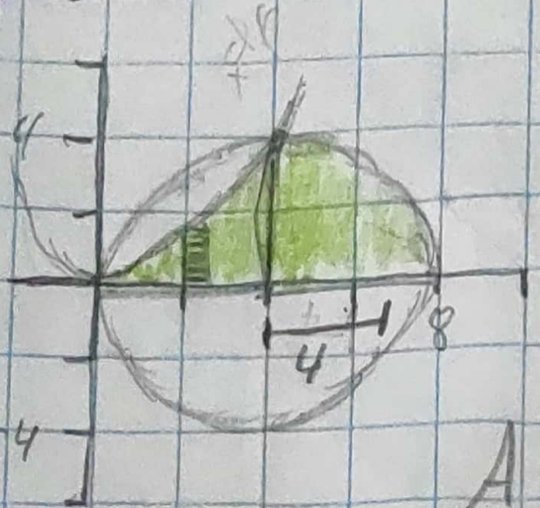
Hoja de Trabajo 2

Ejercicio 1



$$A = \pi r^2 = \frac{\pi r^2}{4} = \frac{\pi 4^2}{4} = \frac{16\pi}{4} = 4\pi$$

$$y = \frac{x^2}{4}$$

$$A = \int_0^4 \frac{x^2}{4}dx = \frac{x^3}{12}\Big|_0^4 = \frac{4^3}{12} = \frac{16}{3}$$

$$A = \frac{16}{3} + 4\pi$$

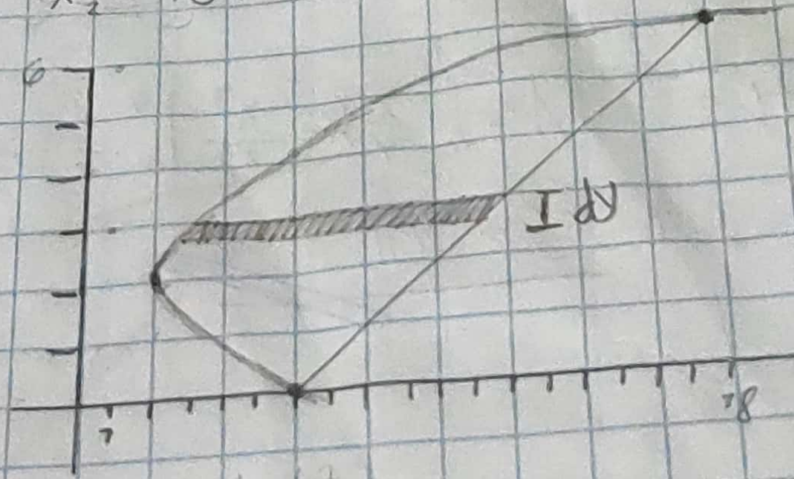
$$y = (x-4)^2 + y^2 = 16$$

Ejercicio 2

$x_1 = y^2 - 4y + 6$

$2y = x - 6$

$x_2 = 2y + 6$



Int

$x_1 = x_2$

$y^2 - 4y + 6 = 2y + 6$

$y^2 - 4y + 6 - 2y - 6 = 0$

$y^2 - 6y = 0$

$y(y - 6) = 0$

$y = 0 \; , \; x = 6$

$y = 6 \; , \; x = 18$ $\quad V =$ area · grosor

$A = l^2 = l = l = x_1 - x_2 = y^2 - 4y + 6 - 2y - 6$

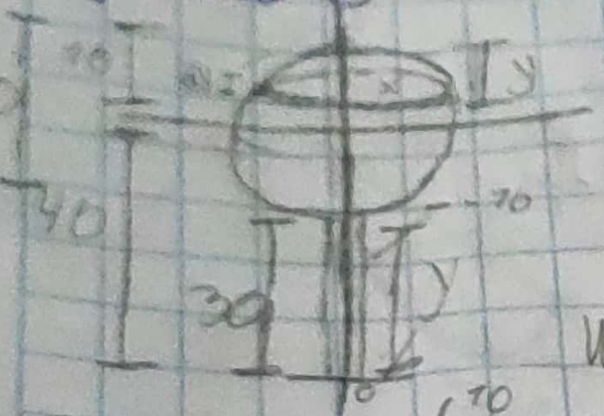
$$V = \int_0^6 (y^2 - 4y + 6 - 2y - 6)^2 \, dy$$

$$V = \int_0^6 (y^2 - 6y)^2 \, dy = \int_0^6 y^4 - 12y^3 + 36y^2 \, dy$$

$$V = \frac{y^5}{5} - 3y^4 + 12y^3 \Big|_0^6$$

$$V = \frac{6^5}{5} - 3(6)^4 + 12(6)^3 = \frac{1296}{5}$$

Ejercicio 3



$W = \rho g\, d\, v$

$C = x^2 + y^2 = r^2$

$x^2 + y^2 = 10^2$

$x^2 + y^2 = 100$

$x = \sqrt{-y^2 + 100}$

$\rho g = 62.4$

$d = 40 + y$

$V = \pi r^2 = \pi\left(\sqrt{-y^2+100}\right)^2$

$V = \pi(-x^2 + 100)$

$$W = 62.4\,(40+y)\left(\pi(-y^2+100)\right)dy$$

$$W = 62.4\pi \int_{-10}^{10} (40+y)(-y^2+100)\,dy$$

$$= 62.4\pi \int_{-10}^{10} -40y^2 + 4000 - y^3 + 100y\,dy$$

$$= 62.4\pi\left(-\frac{40}{3}y^3 + 4000y - \frac{1}{4}y^4 + 50y^2\right)\Big|_{-10}^{10} =$$

$$= 1820000 - (-1508000) = \underline{3328000\pi}$$

$W = \rho g\, d\, v \qquad \rho g = 62.4 \qquad d = y \qquad v = \pi r^2 = \pi\, 0.5^2 = \frac{1}{4}\pi$

$$W = \int_0^{30} 62.4\,(y)\left(\frac{1}{4}\pi\right)dy = 62.4\pi \int_0^{30} \frac{1}{4}y\,dy = 62.4\pi\left(\frac{1}{8}y^2\right)\Big|_0^{30}$$

$$W = \underline{7020\pi}$$

Ejercicio 4 -

$y = x^2 - 8x + 20$, $y = x + 2$, $x = 1$

Ind.

$y_1 = y_2$

$x^2 - 8x + 20 = x + 2$

$x^2 - 8x + 20 - x - 2 = 0$

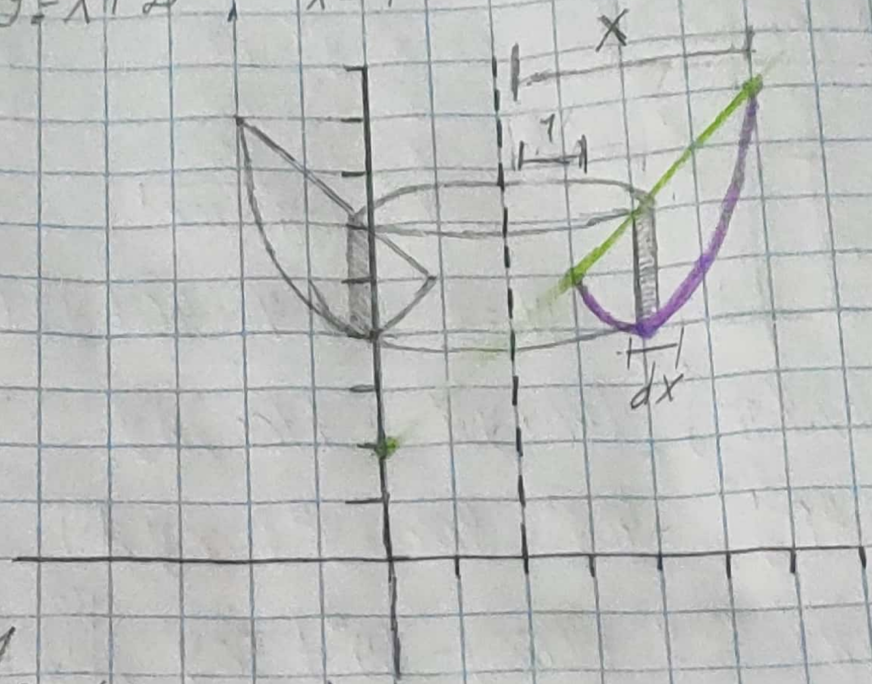
$x^2 - 9x + 18 = 0$

$x(x-3) - 6(x-3) = 0$

$(x-3)(x-6) = 0$

$x = 3 , y = 5$

$x = 6 , y = 8$



$V = 2\pi r h \, dx$

$r = x - 1$

$h = (x+2) - (x^2 - 8x + 20)$

$h = x + 2 - x^2 + 8x - 20$

$h = 9x - 18 - x^2$

$V = 2\pi \int_3^6 (x-1)(9x - 18 - x^2)dx = 9x^2 - 18x - x^3 - 9x + 18 + x^2$

$V = 2\pi \int_3^6 (9x^2 - 18x - x^3 - 9x + 18 + x^2 dx = 2\pi \left(3x^3 - 9x^2 - \frac{x^4}{4} - \frac{9}{2}x^2 + 18x + \frac{x^3}{3}\right)\Big|_3^6$

$V = 2\pi\left[\left(3(6)^3 - 9(6)^2 - \frac{6^4}{4} - \frac{9}{2}(6)^2 + 18(6) + \frac{6^3}{3}\right) - \left(3(3)^3 - 9(3)^2 - \frac{3^4}{4} - \frac{9}{2}(3)^2 + 18(3) + \frac{3^3}{3}\right)\right]$

$V = 36\pi - \frac{9}{2}\pi = \frac{63}{2}\pi \, u^3$